Persistence of Entropy–Information Scaling in Three Dimensions

Breakdown of a Simple Exponent Trend

in Nonlinear Field Systems

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Abstract

The Critical Coherence Index (CCI) has recently been shown to be consistent with an empirical entropy–information scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}$$

in nonlinear field systems [3, 4]. Previous studies established this relation in one- and two-dimensional ϕ^4 dynamics, with an apparent increase of the effective exponent α with spatial dimension.

In this work we extend the analysis to three dimensions. We find that the entropy–information scaling persists in a strong monotonic sense, with Spearman correlation $r_s \approx 0.85$ ($p \approx 1.6 \times 10^{-7}$). However, the notion of a single effective exponent breaks down: instead of a clear optimum, a broad plateau of near-maximal correlation appears over the range $\alpha \in [2.8, 3.5]$. A log-linear regression shows significantly reduced explanatory power ($R^2 \approx 0.37$) compared to lower dimensions.

These results indicate that the entropy–information scaling remains robust across dimensions, while its quantitative structure undergoes a dimension-dependent restructuring. Rather than a simple monotonic power-law trend $\alpha(d)$, the system transitions toward a broad effective scaling regime with a degeneracy of effective exponents.

All results are empirical; a first-principles derivation remains an open problem.

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1 Introduction

Understanding which structures persist in nonlinear field systems remains a central problem in nonequilibrium dynamics. The Critical Coherence Index (CCI) was introduced as a diagnostic of structural instability [2], quantifying the balance between destabilising activity and coherence-preserving structure.

In one-dimensional ϕ^4 dynamics, the CCI was found to be consistent with an empirical scaling relation involving coarse-grained histogram activity and structural mutual information [3], with $\alpha \approx 2.5$. In two dimensions, the scaling persists with a shifted exponent $\alpha \gtrsim 3.0$ and stronger correlation [4]. A comparative analysis [5] suggested a possible dimensional trend $\alpha_{\text{eff}}(d)$.

The goal of the present work is to test this trend in three dimensions and investigate whether the scaling structure persists.

2 Model and Method

We consider a three-dimensional ϕ^4 scalar field:

$$\ddot{\phi} = \nabla^2\phi - (\phi^3 - \phi), \tag{1}$$

on a periodic cubic $24 \times 24 \times 24$ lattice ($\Delta x = 1$), integrated with a fourth-order Runge–Kutta scheme ($\Delta t = 0.02$, $T_{\max} = 8.0$, seed $= 42$).

Initial conditions are random mixtures of vacuum states, domain-wall-like structures (with random orientation), localised Gaussian excitations, and additive noise.

For each of the $N = 24$ trajectories we compute:

- the Critical Coherence Index $\mathcal{C}_{\text{CCI}}$,
- the histogram activity proxy $\dot{S}_{\text{cg}}^{(+)}$,
- the nearest-neighbour mutual information I_{nn}, averaged over all three lattice directions.

The scaling ansatz $R_\alpha = \dot{S}_{\text{cg}}^{(+)}/I_{\text{nn}}^{\alpha}$ is tested by scanning $\alpha \in [0.5, 3.5]$ and maximising the Spearman rank correlation between $\mathcal{C}_{\text{CCI}}$ and R_α. All code is available at [6].

3 Results

3.1 Three-Dimensional Scaling

The entropy–information scaling persists in three dimensions. The maximum Spearman correlation is:

$$r_s \approx 0.85, \qquad p \approx 1.6 \times 10^{-7}, \tag{2}$$

confirming a strong monotonic relationship.

However, unlike in 1D and 2D, the dependence on α is not sharply peaked. Instead, a broad plateau of near-maximal correlation appears over the range:

$$\alpha \in [2.8,\ 3.5] \quad \text{(within tested range)}, \tag{3}$$

with essentially identical Spearman correlations throughout. The notion of a single effective exponent breaks down: the system exhibits a degeneracy of effective scaling exponents.

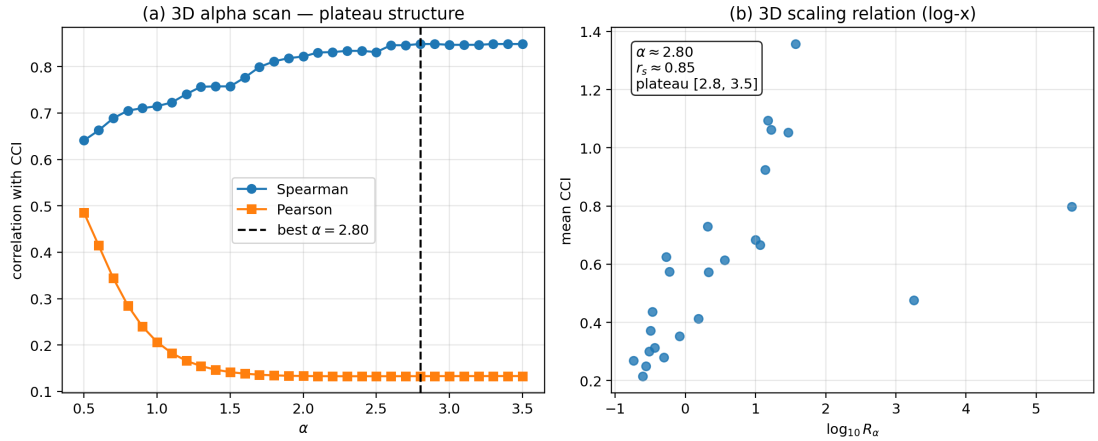

Figure 1: Three-dimensional entropy–information scaling of the CCI. *(a)* Correlation between $\mathcal{C}_{\mathrm{CCI}}$ and $R_\alpha = \dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ as a function of α. Unlike 1D and 2D, the curve shows a broad plateau over $\alpha \in [2.8, 3.5]$ rather than a sharp optimum, indicating a degeneracy of effective scaling exponents. *(b)* Scaling relation between $\mathcal{C}_{\mathrm{CCI}}$ and the best ratio R_α on a log-x axis, illustrating the monotonic dependence underlying the high Spearman correlation ($r_s \approx 0.85$).

3.2 Log-Scaling Fit

A log-linear regression of the form

$$\ln(\mathcal{C}_{\mathrm{CCI}}) = a_0 + a_1 \ln(\dot{S}_{\mathrm{cg}}^{(+)}) + a_2(-\ln I_{\mathrm{nn}}) \tag{4}$$

yields:

$$a_1 \approx 0.32, \quad a_2 \approx 0.27, \quad R^2 \approx 0.37. \tag{5}$$

In contrast to 2D, where structural information dominates ($a_2 \approx 0.49 \gg a_1 \approx -0.08$), the 3D system shows a more balanced contribution from histogram activity and information loss, but with significantly reduced predictive power.

3.3 Comparison Across Dimensions

Table 1 summarises the scaling behaviour across spatial dimensions. Figures 1 and 2 show the corresponding plots.

Table 1: Entropy–information scaling across spatial dimensions. The 2D exponent is a lower bound within the tested range; the 3D system exhibits a plateau rather than a single optimum.

Dimension	α_{eff}	r_s	R^2	Dominant
1D	≈ 2.5	0.73	moderate	/I balance
2D	$\gtrsim 3.0$	0.87	0.74	I_{nn} loss
3D	plateau [2.8, 3.5]	0.85	0.37	balanced

4 Discussion

4.1 Breakdown of a Simple Power-Law Trend

The results do not support a simple monotonic increase of α with spatial dimension. While the Spearman correlation strengthens from 1D to 2D, the 3D system exhibits a plateau of degenerate exponents rather than a sharpened optimum.

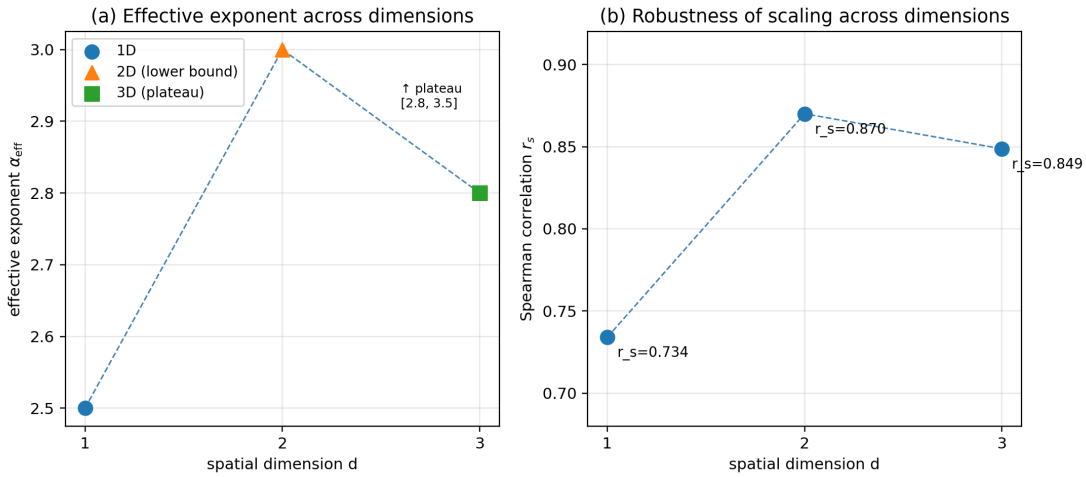

Figure 2: Dimensional comparison of entropy–information scaling. *(a)* Effective exponent α_{eff} across dimensions. The 2D value is a lower bound within the tested range; the 3D system exhibits a plateau rather than a single optimum. *(b)* Spearman correlation r_s across dimensions, showing that the scaling relation remains strong (and even strengthens from 1D to 2D), but does not continue to increase in 3D.

This suggests that the notion of a single effective exponent becomes increasingly less meaningful in higher dimensions, leading to a degeneracy of effective scaling exponents.

4.2 Persistence of the Scaling Structure

Despite the breakdown of a sharp power-law, the strong Spearman correlation ($r_s \approx 0.85$) confirms that the CCI continues to be governed by a balance between histogram activity and structural information. The scaling structure is robust; only its precise quantitative form changes with dimension.

4.3 Interpretation

A consistent picture emerges across dimensions:

- **1D:** clear scaling with a moderate, well-defined exponent.
- **2D:** strong scaling with a dominant information contribution and a shifted exponent.
- **3D:** persistent scaling but with a broad exponent plateau, indicating a degeneracy of effective scaling behaviours.

This suggests that higher-dimensional systems introduce additional degrees of freedom that lead to a degeneracy of effective scaling exponents, rather than a simple continuation of the lower-dimensional trend.

4.4 Relation to the Structural Free Energy

Within the broader framework [2], the entropy–information ratio serves as an empirical projection of the structural free energy onto a lower-dimensional scaling plane. The emergence of a broad plateau in 3D may reflect an increasing number of relevant projection directions in higher dimensions, making a single scalar exponent insufficient to characterise the scaling structure.

5 Conclusion

We have extended the entropy–information scaling analysis of the CCI to three-dimensional ϕ^4 field dynamics. The scaling persists with strong Spearman correlation ($r_s \approx 0.85$, $p \approx 1.6 \times 10^{-7}$), confirming its robustness across dimensions.

However, the simple power-law trend observed from 1D to 2D does not continue: the 3D system exhibits a broad plateau of effective exponents $\alpha \in [2.8, 3.5]$, indicating a degeneracy of scaling descriptions.

These results suggest that the entropy–information scaling of the CCI undergoes a dimension-dependent restructuring: a transition from a well-defined power-law exponent in low dimensions to a broad effective scaling regime in higher dimensions.

Future work will explore:

- higher-resolution 3D simulations,
- extension to other field models,
- a theoretical derivation of the plateau structure and its connection to the structural free energy framework.

All results are empirical. A first-principles derivation of the scaling structure and its dimensional dependence remains an open problem.

A Reproducibility

All code, data, and figures for this paper are available at:

https://github.com/Chris4081/structural-selection-principle

Script for this paper:

- `cci_entropy_scaling_3d.py` — 3D ensemble simulation, α-scan, log-linear fit, and all figures.

Output files:

- `3d_cci_entropy_information_test.csv` — ensemble results
- `3d_cci_alpha_scan.csv` — correlation scan over α
- `3d_cci_alpha_scan.png` — intermediate: α-scan
- `3d_cci_vs_best_ratio_logx.png` — Figure: CCI vs ratio (log-x)
- `paper11_figure1_3d_scaling.png` — Figure 1: 3D scaling summary
- `paper11_dimension_comparison.png` — Figure 2: dimensional comparison

Run:

```
pip install numpy pandas matplotlib scipy scikit-learn
python cci_entropy_scaling_3d.py
```

Key parameters: $N = 24 \times 24 \times 24$, $\Delta t = 0.02$, $T_{\max} = 8.0$, seed $= 42$, $N_{\text{runs}} = 24$, $\alpha \in [0.5, 3.5]$ (31 steps).

References

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